



RAOH

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Research Ethics & Institutional Review Board (IRB)

Protecting Human Participants in Medical Research

Every medical discovery begins with a question.

Researchers investigate diseases, treatments, behaviors, and health outcomes to improve patient care and advance scientific knowledge. However, scientific progress should never come at the expense of human dignity, safety, or rights.

This fundamental principle gave rise to what we now call **Research Ethics**.

Research ethics ensures that participants are treated with respect, protected from unnecessary harm, and fully informed before agreeing to take part in a study.

In modern medicine, no research involving human participants should begin without ethical oversight and approval.

What Is Research Ethics?

Research Ethics refers to the moral principles and professional standards that guide the design, conduct, and reporting of scientific research involving human participants.

Its primary purpose is simple:

To advance scientific knowledge while protecting the rights, welfare, privacy, and dignity of participants.

Research ethics applies to all forms of human research, including:

- Surveys and questionnaires
- Clinical trials
- Interviews
- Medical record reviews
- Biological sample collection
- Behavioral and psychological studies

Why Do We Need Research Ethics?

Researchers often have access to highly sensitive information and may interact directly with participants.

Without ethical safeguards, research can lead to:

- Physical harm
- Psychological distress
- Loss of privacy

- Exploitation of vulnerable groups
- Misuse of personal data

Research ethics exists to prevent these outcomes and maintain public trust in science.

The Four Fundamental Principles of Research Ethics

1. Autonomy

Respect for Individual Choice

Every participant has the right to make their own decisions regarding participation.

No person should be forced, pressured, or manipulated into joining a study.

Participants must be provided with sufficient information to make an informed decision.

Example

A medical student is invited to complete a research questionnaire.

The student must be informed that participation is entirely voluntary and that declining will not affect academic standing in any way.

2. Beneficence

Maximizing Benefits

Research should provide meaningful scientific, educational, or clinical value.

Investigators should design studies that have the potential to improve knowledge, healthcare, or patient outcomes.

Key Question

Does this study provide a genuine benefit that justifies involving human participants?

3. Non-Maleficence

Do No Harm

One of the oldest principles in medicine states:

“First, do no harm.”

Researchers must minimize risks and avoid exposing participants to unnecessary physical, emotional, social, or legal harm.

Examples of Unacceptable Practice

- Exposing participants to unnecessary danger
 - Administering unsafe interventions without justification
 - Collecting sensitive information without adequate protection
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4. Justice

Fairness and Equity

The benefits and burdens of research should be distributed fairly.

Researchers should avoid targeting specific populations simply because they are easier to recruit or less able to refuse participation.

Example

It would be unethical to conduct all high-risk research exclusively on economically disadvantaged populations while other groups receive the benefits.

Informed Consent

The Cornerstone of Ethical Research

One of the most important requirements in any human study is informed consent.

Informed consent is not merely a signature on a form.

It is a process through which participants understand:

- The purpose of the study
- What participation involves
- Potential risks
- Expected benefits
- Privacy protections
- Their right to withdraw at any time

Participation must always be voluntary.

Essential Elements of Informed Consent

A proper informed consent document should clearly explain:

Study Purpose

Why is the research being conducted?

Study Procedures

What will participants be asked to do?

Potential Risks

What discomforts or harms might occur?

Potential Benefits

Will participants or society gain from the research?

Confidentiality

How will personal information be protected?

Voluntary Participation

Can participants refuse or withdraw?

The answer must always be:

Yes, at any time, without penalty.

Confidentiality and Privacy

Protecting participant information is a fundamental ethical responsibility.

Researchers must ensure that personal data remain secure and accessible only to authorized individuals.

Common strategies include:

- Removing names and identifiers
- Assigning participant codes
- Password-protected databases
- Restricted data access

Whenever possible, researchers should use anonymous or de-identified data.

Understanding Research Risk

Every study carries some level of risk.

Ethical review requires balancing those risks against potential benefits.

Physical Risks

Potential bodily harm or discomfort.

Examples:

- Blood sampling
 - Experimental medications
 - Medical procedures
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Psychological Risks

Emotional distress or discomfort.

Examples:

- Questions about trauma
 - Depression screening
 - Sensitive personal topics
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Social Risks

Potential damage to social relationships or reputation.

Examples:

- Disclosure of private information
 - Exposure of stigmatized conditions
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Legal Risks

Potential legal consequences resulting from disclosed information.

Examples:

- Illegal behaviors
 - Confidential legal matters
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What Is an IRB?

Institutional Review Board

An Institutional Review Board (IRB) is an independent committee responsible for reviewing research involving human participants before the study begins.

Its role is to ensure that ethical principles are respected and participant safety is protected.

Think of the IRB as the ethical gatekeeper of scientific research.

No human research should proceed without appropriate ethical review and approval.

What Does the IRB Evaluate?

Before approving a study, the IRB carefully examines several areas.

Risk Assessment

Are participants exposed to unnecessary risks?

Benefit Assessment

Do potential benefits justify the risks?

Participant Selection

Are participants recruited fairly?

Informed Consent

Is the consent process clear and adequate?

Confidentiality

Will personal information remain protected?

Scientific Validity

Is the study design capable of answering the research question?

Poor science is considered unethical because it exposes participants to risk without generating useful knowledge.

Possible IRB Decisions

After reviewing a submission, the committee may issue one of several decisions.

Approved

The study meets ethical requirements and may begin.

Minor Revisions Required

Small modifications must be completed before approval.

Major Revisions Required

Substantial changes are needed before reconsideration.

Rejected

The study does not meet ethical or scientific standards.

The Research Ethics Approval Process

For most student research projects, the pathway follows a structured sequence.

Step 1

Develop a clear research question.

Step 2

Prepare a complete research proposal.

Step 3

Design study methodology and data collection tools.

Step 4

Prepare the informed consent document.

Step 5

Obtain supervisor approval.

Step 6

Submit the application to the IRB.

Step 7

Respond to reviewer comments if necessary.

Step 8

Receive official IRB approval.

Step 9

Begin data collection.

Important:

Data collection must never start before obtaining ethical approval.

Common Ethical Mistakes in Student Research

- Collecting data before IRB approval
- Missing informed consent
- Poor confidentiality measures
- Coercing participants
- Collecting unnecessary personal information
- Sharing identifiable participant data

Avoiding these mistakes is essential for responsible research conduct.

The Supervisor's Role in IRB Submission

Many new students believe that once they have a research idea, they can directly submit it to the IRB themselves. In reality, this is usually not how the process works in most universities.

As an undergraduate student, research is typically conducted under the supervision of a faculty member (Supervisor). The supervisor is responsible for reviewing the study, ensuring that it is scientifically sound and ethically appropriate before it reaches the ethics committee.

Before submission, several documents are usually prepared:

- Research Proposal (Protocol)
- Informed Consent Form
- Human Subjects Research Forms
- Data Collection Tools (Questionnaire, Interview Form, etc.)
- IRB Application Documents required by the university

The supervisor reviews these documents, requests revisions when necessary, and confirms that the methodology and ethical considerations are appropriate.

In many institutions, the supervisor's signature is required before the application can move forward. After all required documents are completed and approved, the submission is sent to the university's ethics committee (IRB) for formal review.

The process generally follows the sequence below:

Research Idea
↓
Student
↓
Supervisor Review
↓
Research Proposal Preparation
↓
Informed Consent Preparation
↓
Human Subjects Research Documents
↓
Supervisor Approval & Signature
↓
IRB Submission
↓
IRB Review
↓
Approval
↓
Data Collection

For this reason, the supervisor is not merely a person who signs paperwork. The supervisor serves as the primary academic mentor and acts as the first level of scientific and ethical review before the study reaches the IRB.

An important rule that every researcher should remember:

No participant recruitment, questionnaire distribution, interviews, or data collection should begin before receiving official IRB approval.

Quick Summary

Research Ethics

Protects participant rights, safety, and dignity.

Autonomy

Respecting individual choice.

Beneficence

Maximizing benefits.

Non-Maleficence

Avoiding harm.

Justice

Ensuring fairness.

Informed Consent

Voluntary and informed participation.

Confidentiality

Protecting personal information.

IRB

Independent committee that reviews and approves human research.

Golden Rule

No participant should be harmed, exploited, deceived, or pressured in the pursuit of scientific knowledge